

1. VLAN Configuration and Inter-VLAN Routing using Cisco Packet Tracer

AIM:

To configure **Virtual Local Area Networks (VLANs)** on a switch and implement **Inter-VLAN Routing** using a router in Cisco Packet Tracer, so that hosts in different VLANs can communicate with each other.

PROCEDURE:

Step 1: Place the devices

Open Cisco Packet Tracer and place the following devices in the workspace:

- 1 Router (1941)
- 1 Switch (2960)
- 4 PCs

Step 2: Connect the devices

Connect the devices using **Copper Straight-Through** cable as follows:

- PC0 FastEthernet0 → Switch0 Fa0/1
- PC1 FastEthernet0 → Switch0 Fa0/2
- PC2 FastEthernet0 → Switch0 Fa0/3
- PC3 FastEthernet0 → Switch0 Fa0/4
- Router0 GigabitEthernet0/0 → Switch0 Fa0/24

Step 3: Configure IP addresses on PCs

Click each PC → **Desktop** → **IP Configuration** and enter the following:

PC0

- IP Address: **192.168.10.2**
- Subnet Mask: **255.255.255.0**

- Default Gateway: **192.168.10.1**

PC1

- IP Address: **192.168.10.3**
- Subnet Mask: **255.255.255.0**
- Default Gateway: **192.168.10.1**

PC2

- IP Address: **192.168.20.2**
- Subnet Mask: **255.255.255.0**
- Default Gateway: **192.168.20.1**

PC3

- IP Address: **192.168.20.3**
- Subnet Mask: **255.255.255.0**
- Default Gateway: **192.168.20.1**

Step 4: Create VLANs on the switch

Open **Switch0** → **CLI** and enter the following commands:

```
enable
configure terminal
vlan 10
name SALES
exit
vlan 20
name HR
exit
```

Step 5: Assign switch ports to VLANs

In the switch CLI, assign ports to the required VLANs:

```
interface range fa0/1-2
switchport mode access
switchport access vlan 10
exit

interface range fa0/3-4
switchport mode access
switchport access vlan 20
exit
```

Step 6: Configure trunk port on the switch

Configure the port connected to the router as trunk:

```
interface fa0/24
switchport mode trunk
exit
end
```

Step 7: Configure the router for Inter-VLAN Routing

Open **Router0** → **CLI** and enter the following commands:

```
enable
configure terminal

interface gigabitEthernet0/0
no shutdown
exit

interface gigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit

interface gigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit

end
```

Step 8: Verify VLAN configuration on switch

Use the following command on the switch:

```
show vlan brief
```

Step 9: Verify trunk configuration

Use the following command on the switch:

```
show interfaces trunk
```

Step 10: Verify router interfaces

Use the following command on the router:

```
show ip interface brief
```

Step 11: Test connectivity using ping

Open **PC0** → **Desktop** → **Command Prompt** and test:

Same VLAN communication:

```
ping 192.168.10.3
```

Ping to gateway:

```
ping 192.168.10.1
```

Inter-VLAN communication:

```
ping 192.168.20.2
```

Also test from **PC2**:

```
ping 192.168.20.3  
ping 192.168.20.1  
ping 192.168.10.2
```

SWITCH CONFIGURATION COMMANDS:

```
enable
configure terminal

vlan 10
name SALES
exit

vlan 20
name HR
exit

interface range fa0/1-2
switchport mode access
switchport access vlan 10
exit

interface range fa0/3-4
switchport mode access
switchport access vlan 20
exit

interface fa0/24
switchport mode trunk
exit

end
```

ROUTER CONFIGURATION COMMANDS:

```
enable
configure terminal

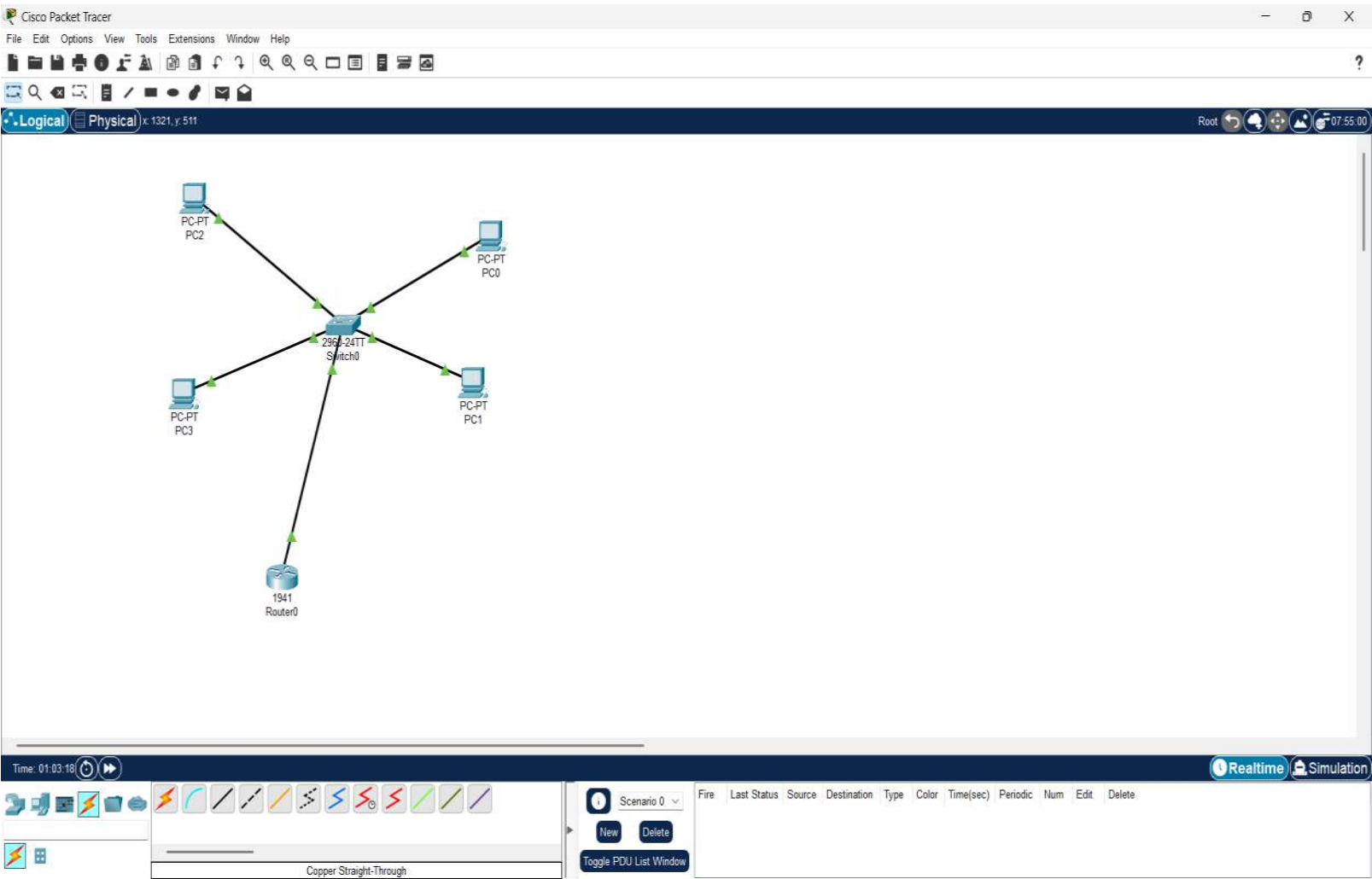
interface gigabitEthernet0/0
no shutdown
exit

interface gigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit

interface gigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit

end
```

OUTPUT:



RESULT:

Thus, VLANs were configured successfully on the switch and Inter-VLAN Routing was implemented using Router-on-a-Stick in Cisco Packet Tracer. Communication between hosts in the same VLAN and in different VLANs was verified successfully using ping commands.